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Task 5 – InfoNCE Loss: Derivation from First Principles  
VLM from Scratch · Seasons of Code · IIT Bombay

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Notation

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|            |                                                                                                                                                |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| N          | number of image-caption pairs in the batch                                                                                                     |
| $I_i, T_j$ | L2-normalized image / text embeddings (unit vectors in $R^D$ )                                                                                 |
| $s_{ij}$   | cosine similarity = $I_i \cdot T_j$ (dot product on the unit sphere)<br>so $s_{ij}$ in $[-1, 1]$ , and $s_{ii}$ is the correct (positive) pair |
| $\tau$     | temperature $> 0$ . We use logits $z_{ij} = s_{ij} / \tau$ .                                                                                   |

Row  $i$  of the similarity matrix holds the similarities of image  $i$  to ALL  $N$  texts.  
Pair  $i$  means image  $i$  should match text  $i$ , so the target for row  $i$  is the index  $i$  (the diagonal).

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1. InfoNCE for one direction (image  $\rightarrow$  text) as an explicit sum

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Principle: "maximize the log-probability of the correct pair under a softmax over all candidate texts." For image  $i$  we define a probability distribution over the  $N$  candidate texts using the scaled similarities as logits:

$$p_i = \frac{\exp(s_{ii} / \tau)}{\sum_{k=1..N} \exp(s_{ik} / \tau)} \quad (\text{prob. that } i\text{'s match is text } i)$$

We want  $p_i$  to be large for every  $i$ . Maximizing the average log-probability is the same as minimizing its negative – this negative-log-likelihood IS the loss:

$$\begin{aligned} L_{i2t} &= - (1/N) * \sum_{i=1..N} \log p_i \\ &= - (1/N) * \sum_{i=1..N} \log \left[ \frac{\exp(s_{ii}/\tau)}{\sum_{k=1..N} \exp(s_{ik}/\tau)} \right] \\ &= - (1/N) * \sum_{i=1..N} \left[ \frac{s_{ii}}{\tau} - \log \sum_{k=1..N} \exp(s_{ik}/\tau) \right] \quad (*) \end{aligned}$$

This is exactly cross-entropy applied row-wise to the matrix  $z = s/\tau$  with target labels  $(1,2,...,N)$  on the diagonal. The full symmetric CLIP loss adds the text  $\rightarrow$  image direction (softmax down each COLUMN) and averages:

$$L = (1/2) ( L_{i2t} + L_{t2i} ).$$

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2. Limiting behavior in the temperature  $\tau$

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Look at one term of (\*):  $L_i = -s_{ii}/\tau + \log \sum_k \exp(s_{ik}/\tau)$ .

(a)  $\tau \rightarrow 0$  (sharp). Factor out  $1/\tau$ . The log-sum-exp becomes a hard maximum:

$$\log \sum_k \exp(s_{ik}/\tau) \rightarrow (1/\tau) * \max_k s_{ik} \quad \text{as } \tau \rightarrow 0,$$

because the largest term dominates the sum exponentially. Hence

$$L_i \rightarrow (\max_k s_{ik} - s_{ii}) / \tau.$$

- If  $s_{ii}$  is the strict maximum of row  $i$ , the bracket is 0 and  $L_i \rightarrow 0$ .
- If ANY off-diagonal  $s_{ij} > s_{ii}$ , the bracket is a positive constant divided by  $\tau \rightarrow 0+$ , so  $L_i \rightarrow +\infty$ .

So as  $\tau \rightarrow 0$  the loss becomes an all-or-nothing constraint: "the correct pair's similarity must be strictly greater than every negative's." Even a tiny violation is punished without bound. This is max-margin / hard-negative behavior – the model becomes obsessed with the single hardest negative in each row.

(b)  $\tau \rightarrow \infty$  (flat). Now  $z_{ik} = s_{ik}/\tau \rightarrow 0$  for all  $k$ , so every  $\exp(\dots) \rightarrow 1$  and the softmax becomes uniform:  $p_i \rightarrow 1/N$ . Therefore

$$L_i \rightarrow -\log(1/N) = \log N, \quad \text{a constant independent of the embeddings.}$$

Since  $L$  no longer depends on  $s_{ij}$ , its gradient w.r.t. every similarity  $\rightarrow 0$ . Training stalls: there is no signal telling embeddings to move. This is the "logits are flat, gradients vanish" failure mode. The loss has collapsed to the fixed value  $\log N$  – the same baseline we see for random embeddings.

Intuition: temperature sets how sharply the model distinguishes the positive from the negatives. Too sharp  $\rightarrow$  unstable, dominated by one hard negative. Too flat  $\rightarrow$  no distinction, no learning. CLIP therefore LEARNS  $\tau$  (as  $\log(1/\tau)$ , clamped).

### ----- 3. Gradients w.r.t. the similarities, and their signs -----

Take one row's term  $L_i = -\log p_i$  with logits  $z_{ik} = s_{ik}/\tau$ . This is a plain softmax cross-entropy with target index  $i$ , so its gradient w.r.t. the logits is the standard "softmax minus one-hot":

$$dL_i / dz_{ij} = p_{ij} - [j == i]$$

where  $p_{ij} = \text{softmax}_k(z_{ik})_j = \exp(s_{ij}/\tau) / \sum_k \exp(s_{ik}/\tau)$ , and  $[j == i]$  is 1 on the diagonal, 0 otherwise. Chain rule through  $z = s/\tau$  ( $dz_{ij}/ds_{ij} = 1/\tau$ ) and the overall  $1/N$  average give:

Correct pair ( $j = i$ ):

$$dL_{i2t} / ds_{ii} = (1 / (N * \tau)) * (p_{ii} - 1) \leq 0 \quad (\text{since } p_{ii} \leq 1)$$

Incorrect pair ( $j \neq i$ ):

$$dL_{i2t} / ds_{ij} = (1 / (N * \tau)) * p_{ij} \geq 0$$

(since  $p_{ij} \geq 0$ )

Interpretation of the signs (gradient DESCENT moves opposite the gradient):

- $ds_{ii}$  gradient is negative  $\Rightarrow$  descent INCREASES  $s_{ii}$ . The correct image-caption pair is PULLED TOGETHER. The pull is strongest when  $p_{ii}$  is small, i.e. exactly when the model is currently wrong – the update is largest where it is most needed, and fades to zero once  $p_{ii} \rightarrow 1$ .
- $ds_{ij}$  gradient is positive  $\Rightarrow$  descent DECREASES  $s_{ij}$ . Every incorrect (image  $i$ , text  $j$ ) pair is PUSHED APART. The push on a negative is proportional to  $p_{ij}$ : hard negatives (those the model wrongly finds similar, large  $p_{ij}$ ) are repelled hardest; easy negatives are barely touched.
- Conservation check:  $\sum_j dL_i/dz_{ij} = (\sum_j p_{ij}) - 1 = 1 - 1 = 0$ . The total pull on the positive exactly balances the total push on the negatives, so the embeddings are rearranged on the sphere rather than all collapsing or exploding. The  $1/\tau$  prefactor shows temperature also scales gradient MAGNITUDE: smaller  $\tau \rightarrow$  larger gradients (another reason tiny  $\tau$  is unstable).

The text $\rightarrow$ image direction gives the symmetric statements down each column; averaging the two directions yields the final CLIP gradient.

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Summary: InfoNCE = row/column softmax cross-entropy on cosine
similarities scaled by  $1/\tau$ , targeting the diagonal. It pulls
positives together, pushes negatives apart with force proportional
to how wrongly-similar they are, and needs a well-chosen (learned,
clamped) temperature to avoid the  $\tau \rightarrow 0$  instability and the
 $\tau \rightarrow \infty$  collapse to the constant  $\log N$ .
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